

## Molecular generator



### SAFE-GPT — autoregressive

Prompt **C C = 7 .**

Step 1 **C C = 7 . c**

Step 2 **C C = 7 . c 1**

⋮

Final **C C = 7 . c 1 5 c c c c ...**



### GenMol — masked diffusion

Step 0 **C C = 7 .** [Masked]

Step 1 **C C = 7 . c** [Masked]

Step 2 **C C = 7 . c 1 5** [Masked]

⋮

Final **C C = 7 . c 1 5 c c c c ...**



### InVirtuoGen — discrete flow

Step 0 **C C = 7 .** [Flow refine] **O N F 8 S o #**

Step 1 **C C = 7 . c** [Flow refine] **N 5 8 S o #**

Step 2 **C C = 7 . c 1 5 c** [Flow refine] **S o #**

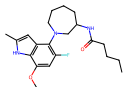
⋮

Final **C C = 7 . c 1 5 c c c c ...**

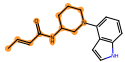
Anchor Generated Flow refine Masked

## EW-SFT

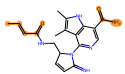
### Sampling



De novo design



Motif extension



Linker design

### Genetic search



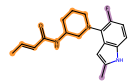
Genetic operator



Population  $P_t$



Attach & Regenerate



Evolved molecules

### Elite-weighted native-loss update

$$\mathcal{L}_t = \sum_{x \in \tilde{B}_t} \bar{w}(x) \ell_{\theta}(x), \quad \bar{w}(x) = \frac{w(x)}{\sum_{x'} |w(x')|}$$

Elite-weighted  
SFT loss

Valid unique  
batch  $\tilde{B}_t$



Oracle scoring  
ROSHAMBO2  
PMO

Update pool  $S_t$

$\tau_t$

Elite/batch  
selection

Selected set

$w$

Within-set  
emphasis